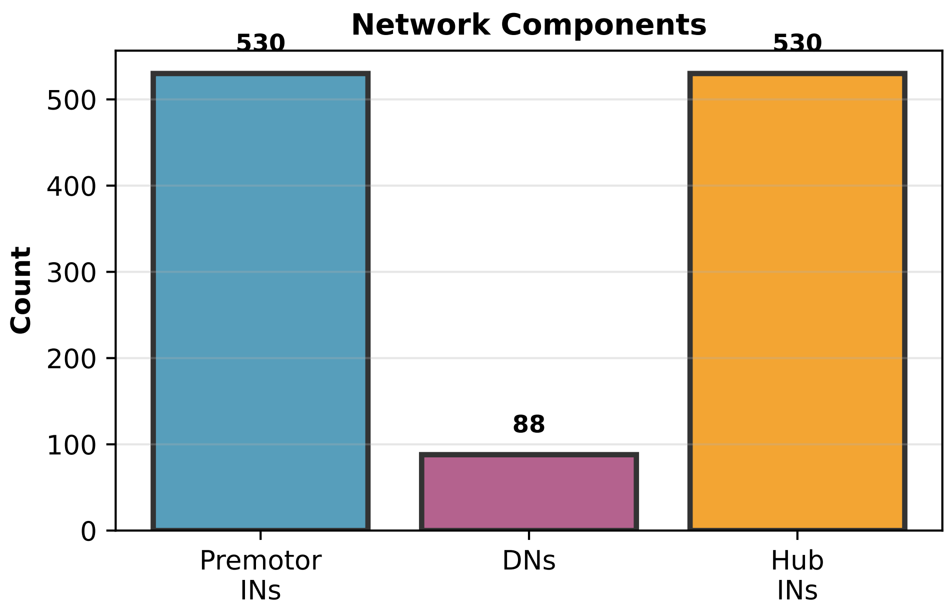
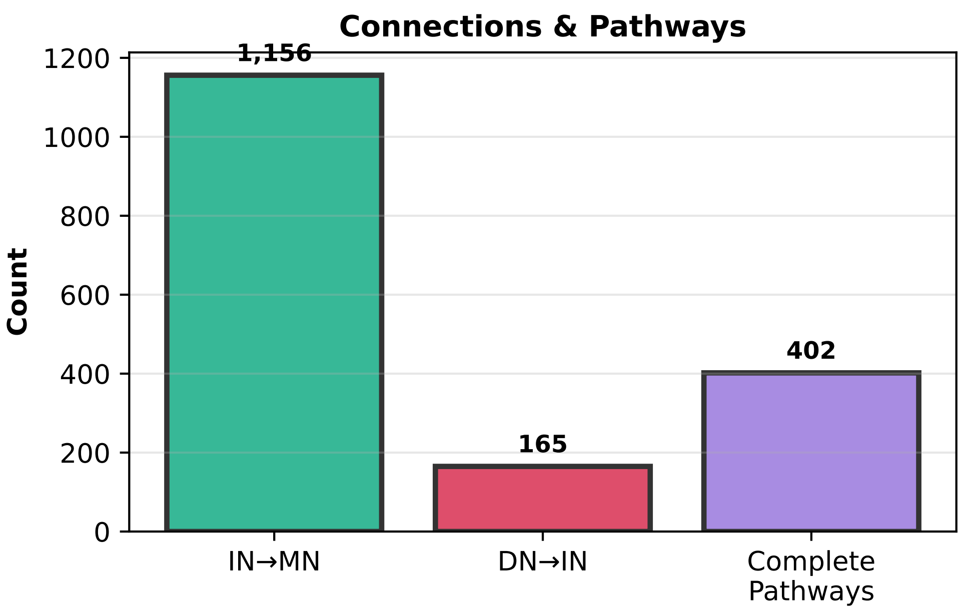
Wing Motor Control Circuit - Complete Analysis Overview



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WING MOTOR CONTROL CIRCUIT ANALYSIS
Complete Network Characterization (min_synapses = 25)
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NETWORK SIZE:
    • Premotor Interneurons:      530
    • IN→MN Connections:          1,156
    • Total IN→MN Synapses:       49,863

    • Descending Neurons:         88
    • DN→IN Connections:          165
    • Total DN→IN Synapses:       6,460

    • Complete DN→IN→MN Pathways: 402

CLUSTERING ANALYSIS:
    • Number of Clusters:         5
    • Silhouette Score:           0.497
    • Calinski-Harabasz Index:    46.8

    Power Control:
        • Clusters: 4
        • INs: 289

    Steering Control:
        • Clusters: 1
        • INs: 241

HUB ANALYSIS:
    • Hub Neurons:                530
    • Hub Percentage:             100.0%

DN SPECIALIZATION:
    • Power DNs:                  26
    • Steering DNs:               62

TOP CANDIDATE INTERNEURON:
    • ID:                         720575941524114276
    • Name:                       IN06B013
    • Cluster:                    2 (power_control)
    • Hub S                       True
    • Pri                         65.6
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QUALITY METRICS

Clustering:
  Silhouette:    0.497
  CH Index:      46.8

Hub Detection:
  Hubs:          530
  Percentage:    100.0%

Network:
  INs:           530
  DNs:           88
  Pathways:      402
```